Answer Key & Teacher Notes

Session 1 — Integers and Opposites · Session 2 — Integers in Real Situations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Integers and Opposites

TALK PROMPTS

- How does a number line show that two opposite elevations are the same distance from zero?
- Why do adding money and using money up get opposite signs?

PARTNER TASK

Elevator Trip. One partner writes integers for 5 floors down and 5 floors up from the lobby; the other explains what 0 means. Switch roles with Tower and Tunnel: 44 feet above and 44 feet below the ground.

WATCH FOR

- Dropping the negative sign for "below" or "debt" (writing 24 for 24 m below sea level).
- Thinking a negative number is its own opposite (opposite of −18 is −18).
- Writing an integer before deciding what 0 means in the situation.

SESSION 2 · TEACHER NOTES

Integers in Real Situations

TALK PROMPTS

- Two cities can sit the same distance from zero. Why are their temperatures still different?
- How do the two signs show that a 45-dollar refund and a 45-dollar charge cancel each other out?

PARTNER TASK

Refunds and Yards. One partner writes integers for a 45-dollar charge and a 45-dollar refund; the other writes integers for a 9-yard loss and a 9-yard gain. Each explains what a total of 0 means.

WATCH FOR

- Believing two opposites always make a positive (opposite of the opposite of −19 written as 19).
- Using a positive number for a loss, drop, or withdrawal.
- Thinking two readings the same distance from 0 are equal.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.5	1	1	2	1	2	2	2	3	2	1	1
S1 · Form B	6.NS.C.5	1	1	2	1	2	2	2	3	2	1	1
S2 · Form A	6.NS.C.5	1	1	2	1	1	2	2	3	2	1	1
S2 · Form B	6.NS.C.5	1	1	2	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

integer: Zero, a counting number, or the **opposite** of a counting number.

opposite: A number the same **distance** from 0, on the other side of 0.

negative number: A number **less** than 0, written with a − sign.

positive number: A number **greater** than 0.

- First decide what **0** means in the situation.
- Gains, above, and deposits are **positive**. Losses, below, and debts are **negative**.

Example: Zero means the **starting** score. Earning 8 points is **8**. Losing 8 points is **−8**. Each is 8 units from 0, on opposite sides. Answer: 8 and −8 are **opposites**.

YOU TRY

1. a. −14 b. 6

2. Points at −4 and 4

EXIT TICKET

1. a. −20 b. 11

2. Points at −5 and 5; 5 units each

3. **A** (−4 and 4)

PRACTICE FORM A

1. a. 12 b. −25 c. −8 d. −3

2. a. 19 b. −6 c. 0

3. a. 35, −20, 15, −35 b. Ridge Point and Cave Floor (35 and −35)

4. −12; opposite 12

5. No; the opposite of −18 is 18

6. −7, 7, 0; opposite pair: −7 and 7

7. a. −31 ft (windmill 31) b. ground level

8. a. −3 and 7 b. 4 c. −4 means 4 floors below the lobby

9. Below sea level is negative, so the elevation is −24 m. An opposite is on the other side of 0, so the opposite of −24 is 24 (24 m above sea level).

10. **B** (−30)

11. **C** (0)

Self-check word **INTEGER**

PRACTICE FORM B

1. a. 17 b. −32 c. −6 d. −4

2. a. 23 b. −9 c. 0

3. a. 28, −16, 12, −28 b. Lookout and Dive Station (28 and −28)

4. −15; opposite 15

5. No; the opposite of −21 is 21

6. −5, 5, 0; opposite pair: −5 and 5

7. a. −44 ft (tower 44) b. ground level

8. a. −5 and 8 b. 3 c. −3 means 3 floors below the lobby

9. Below sea level is negative, so the elevation is −26 m. An opposite is on the other side of 0, so the opposite of −26 is 26 (26 m above sea level).

10. **B** (9)

11. **C** (15)

Self-check word **NEGATIVE**

GUIDED NOTES

opposite: A number the same distance from 0, on the **other** side of 0.

opposite of the opposite: Flip across 0 **twice** and you are back where you started.

integer: A whole number or its **opposite**.

- The sign shows the **direction** of a change from a starting point.
- −(−9) means the opposite of −9, which is **9**.

Example: Zero means **no change** in the water. Monday's gain is **25**. Tuesday's loss is **−25**. Answer: The two changes are **opposites**: same size, opposite effects.

YOU TRY

1. a. 11 b. 9

2. −14 m (first flip 14, second flip −14)

EXIT TICKET

1. a. −20 b. 13

2. −7; opposite 7, which means 7 degrees above zero

3. **B** (9)

PRACTICE FORM A

1. a. −9 b. 45 c. −6 d. 200

2. a. 16 b. −7 c. 21 d. −8

3. a. points at −15, −8, 8, 21 b. Denver and Oslo c. 8 units

4. 33 degrees

5. −7 and 7 (0 is ground level)

6. No; −26 goes to 26, then back to −26. The value is −26.

7. −30, −11, 0, 30; opposite pair: −30 and 30

8. a. 36 m b. −36 m c. the cave entrance

9. The first flip takes −19 to 19. The second flip goes back across 0 to −19. Two opposites return to the start, so the answer is −19.

10. **B** (45)

11. **B** (−75)

Self-check word **OPPOSITE**

PRACTICE FORM B

1. a. −12 b. 30 c. −4 d. 150

2. a. 24 b. −5 c. 38 d. −6

3. a. points at −13, −6, 6, 19 b. Pine Lodge and Summit Lodge c. 6 units

4. 27 degrees

5. −9 and 9 (0 is ground level)

6. No; −14 goes to 14, then back to −14. The value is −14.

7. −25, −8, 0, 25; opposite pair: −25 and 25

8. a. 42 m b. −42 m c. the cliff top

9. The second opposite must also flip across 0: 22 goes to −22. Two opposites return to the start, so the answer is −22.

10. **B** (52)

11. **B** (−60)

Self-check word **POSITIVE**

SHEET 1 · INTEGERS AND OPPOSITES

1. a. −12 b. 40 c. −9 d. 15 e. −25 f. 7

2. Points at −6 and 6; each is 6 units from 0.

3. A: −7, opposite 7; B: −2, opposite 2; C: 3, opposite −3; D: 9, opposite −9

4. Opposites are on different sides of 0, so the sign changes. −15 is 15 units left of 0; its opposite is 15 units right of 0. The opposite of −15 is 15.

SHEET 2 · INTEGERS IN REAL SITUATIONS

1. a. −35 b. 20 c. −4 d. −8

2. a. 9 b. −15 c. −4

3. A at −3, B at 3, C at −3. A and C are the same point: the opposite of the opposite of −3 is −3.

4. a. noon 6, midnight −6 b. They are opposites: each is 6 units from 0, on different sides.

5. No. The opposite of a negative number is positive. Example: the opposite of −7 is 7, which is positive.

6. **B** (−18)

7. **B** (−(−11) = 11)

October 2026 conceptual additions — teacher key

Session 1 · Integers and Opposites

Changes +8 and −8 L are opposites. Their net change is zero, but the tank need not contain zero liters; zero means no change from the reference level.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Changes +8 and −8 L are opposites. Their net change is zero, but the tank need not contain zero liters; zero means no change from the reference level.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Integers in Real Situations

A balance is a current position; a change reduces the balance by 12. −(−12)=12; its opposite is −12. Context defines the reference and role.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: A balance is a current position; a change reduces the balance by 12. −(−12)=12; its opposite is −12. Context defines the reference and role.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.